Student ID: **Student Name:**

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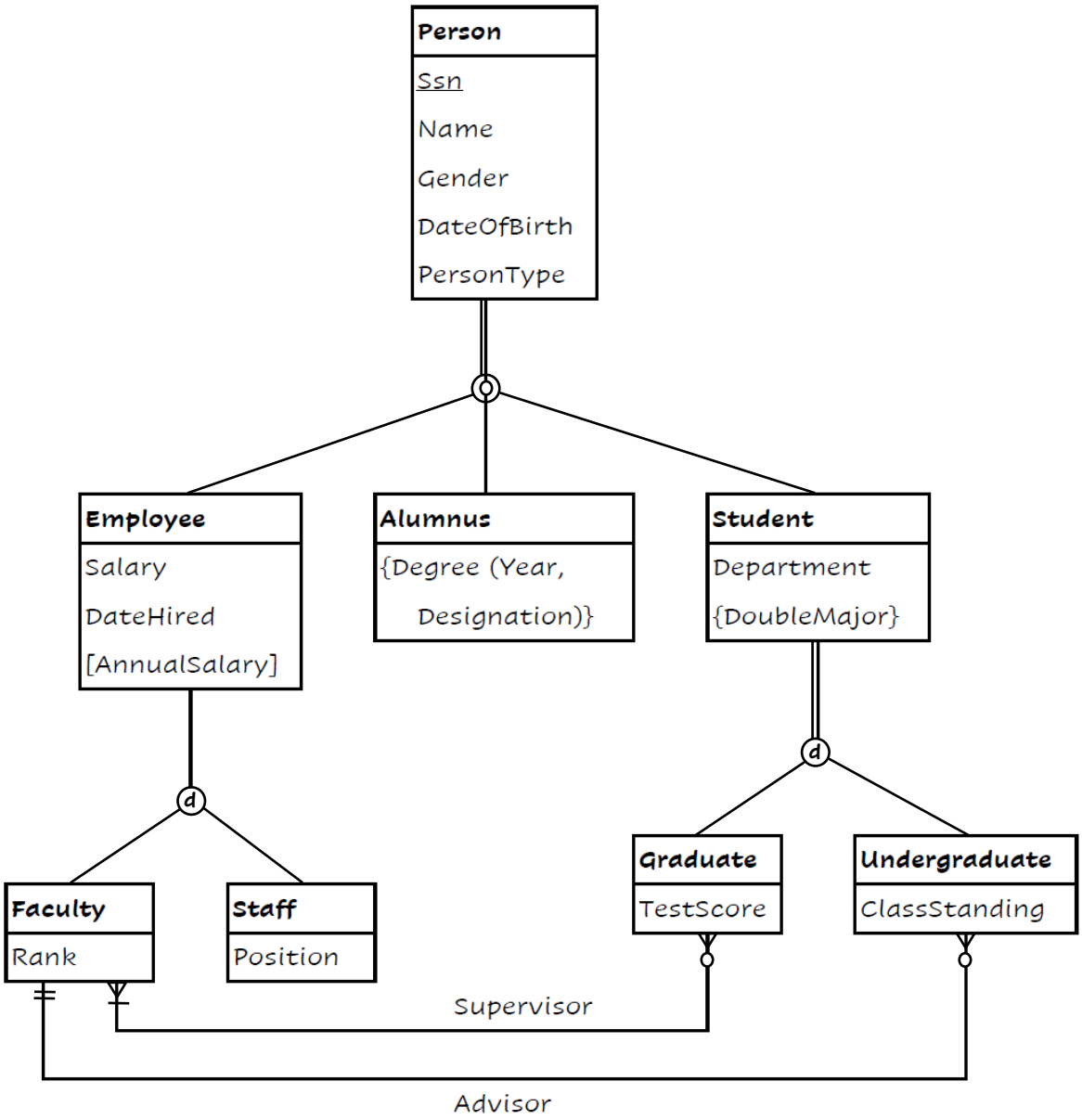
Marmara University – Faculty of Engineering – Department of Computer Engineering

Fall 2020 – CSE3055 Database Systems

Quiz #2

(26.11.2020 | Duration: 45 min)

1) [40 pts] Map the following enhanced entity-relationship diagram into a relational database schema *i.e.* tables and fields. Please state any assumptions that you have made. Do not forget to use arrows for foreign key – primary key references. Also, note that the attribute *PersonType* is a subtype discriminator.



2) [60 pts] Consider the following database. Write SQL statements for the following queries using the relational database below.

ALBUM			
albumID	**title**	**year**	**singerID**
1	Festival	2006	11
2	Patron	2009	11
3	Mançoloji	1999	6
5	Sakin Ol	1992	4
6	Turuncu	2001	4
7	Sade	2013	4

SINGER		
singerID	**name**	**style**
1	Ferhat Göçer	Pop
2	Orhan Gencebay	Arabesk
3	Candan Erçetin	Pop
4	Sertab Erener	Pop
5	Gülben Ergen	Arabesk
6	Barış Manço	Pop
7	Murat Boz	Pop
8	Erol Büyükburç	Pop
9	Demir Demirkan	Rock
10	Zeynep Dizdar	Pop
11	Kenan Doğulu	Pop

INSTRUMENT_SINGER	
singerID	**insID**
1	1
1	3
2	4
3	1
3	8
4	1
4	5
4	6
4	7
11	1
11	3

INSTRUMENT	
insID	**name**
1	Piano
2	Drum kit
3	Zurna
4	Bağlama
5	Guitar
6	Harmonica
7	Trumpet
8	Violin

a) [12 pts] List all the attributes of albums which have a year greater than or equal to 2005.

b) [12 pts] List album title, album year, singer name and singer style of albums whose singer is "Sertab Erener". The order of the list should be descending according to album title.

c) [12 pts] List all the names of instruments played by the singers whose style is "Pop". The list should not display redundant records.

d) [12 pts] List the album ID, title, singer name and style of singers other than the ones who play an instrument such that second character of the instrument's name is "a".

e) [12 pts] List the album title and the name of singers who play at least two different instruments. Remove redundant records in the list.